



Parental care

- Any form of parental behavior that is likely to increase the reproductive potential of offspring

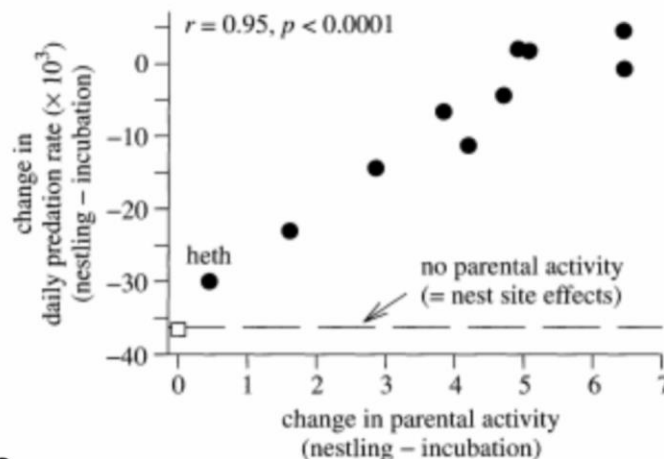
Why parent?

- Parental care is costly!
 - Energetically Expensive
 - Birds: Daily energetic cost of egg laying is around 29-35% of basal metabolic rate, 230% increase in daily protein requirement
 - Reptiles: Egg production $\square$ 20% of annual energy budget



Why parent?

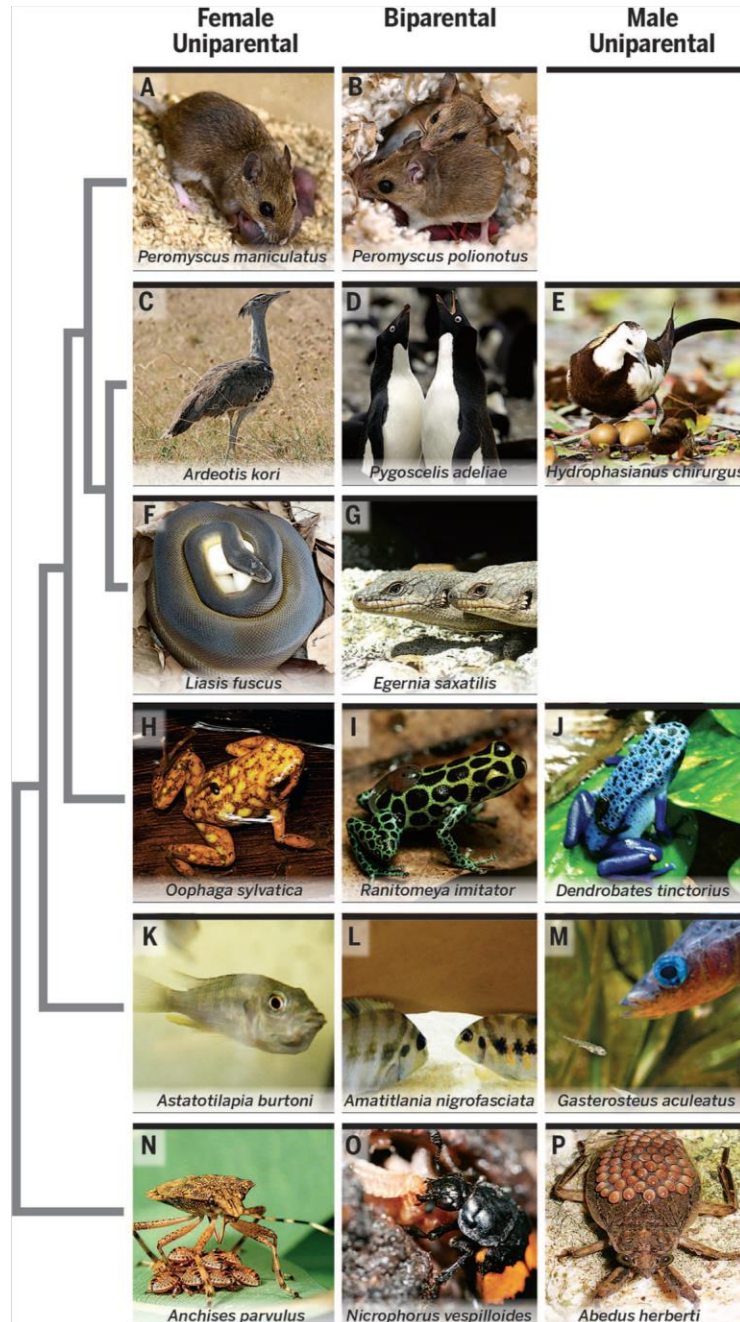
- Parental care is costly!
 - Energetically Expensive:
 - Birds and mammals: Energetic costs of feeding offspring exceed egg production or gestation
 - Lost opportunity costs while providing care
 - Parental care behaviors might attract predators



How common is parental care?

Group	Only Females	Only Males	Both Parents	No Care
Solitary insects	Many	Few	Few	Most
Social Insects	Most	---	Few	---
Reptiles/ Amphibians	Some (internal fertilization)	Some (external fertilization)	---	Most
Fish	Some (internal fertilization)	Many (external fertilization)	Some	Most
Birds	Some	Few	Most	---
Mammals	Most	---	Some	---

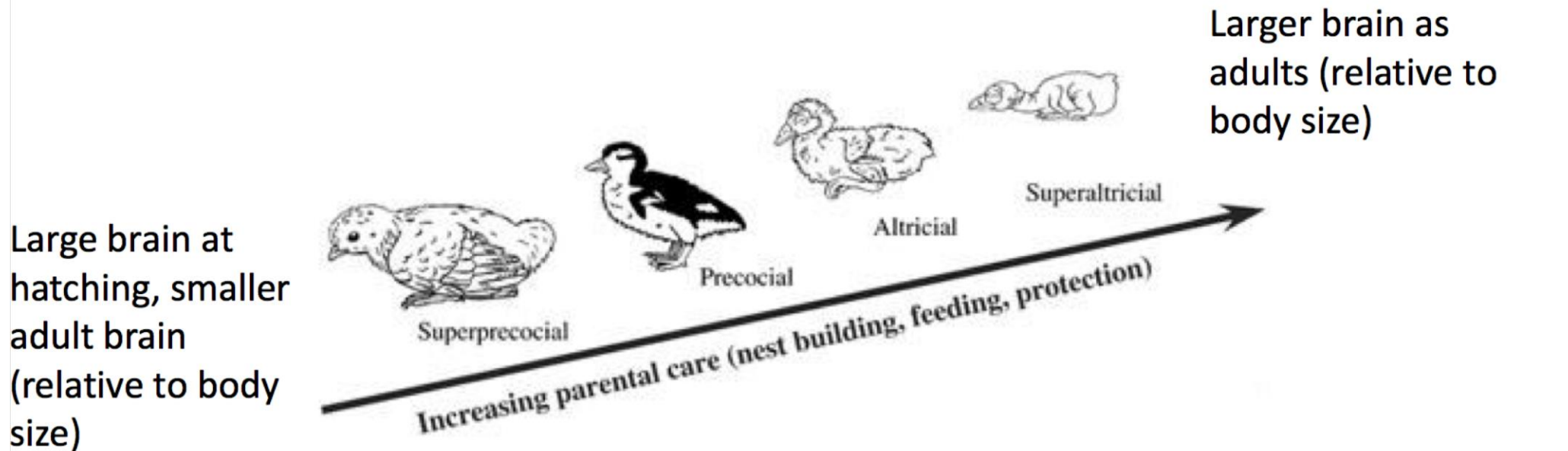
The diversity of parental care



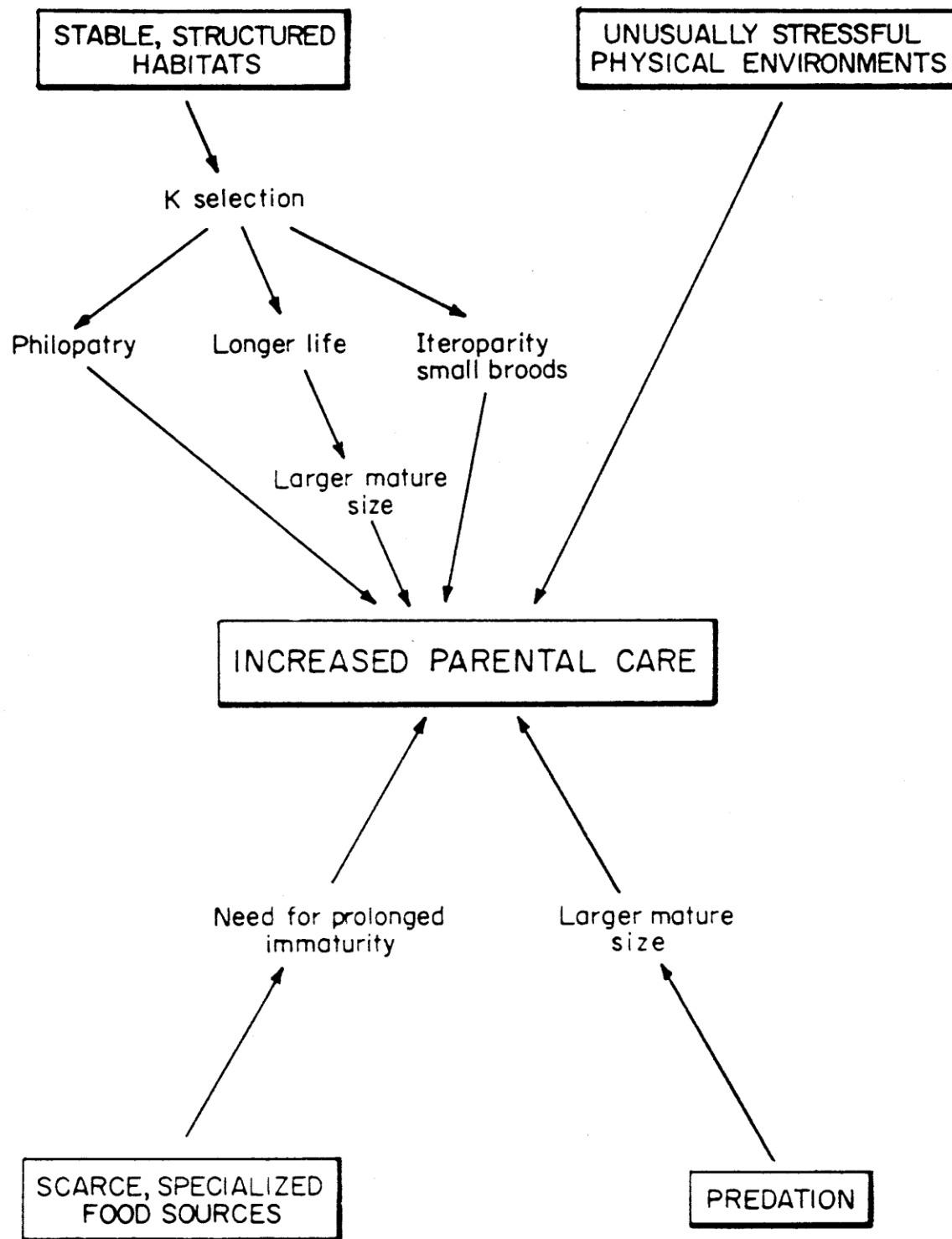
Dulac et al., 2014

How common is parental care?

- In birds, costly parental care has evolved independently multiple times



Ricklefs 1983
Ehrlich et al. 1988
Gill 1995



Who should parent?

- When you can guarantee parentage, parenting is more likely.



Male care is common in fishes that have external fertilization

	Male care		Female care		No care	
Fertilization	Internal	External	Internal	External	Internal	External
Fish	2	61	14	24	5	100
Amphibians	2	14	11	8	0	10
Total	4	75	25	32	5	110

families

Gross and Shine 1981

R and K Strategies

- Sliding scale of life history characteristics.
- R strategies emphasize rapid reproduction, low investment
- K strategies emphasize slow reproduction, high investment.



- R are short-lived, many offspring at once.
- K are long-lived, few offspring at a time.



Infidelity among r-strategists is devastating.

	<i>r</i> Unstable environment, density independent	<i>K</i> Stable environment, density dependent interactions
Organism size	Small	Large
Energy used to make each individual	Low	High
# Offspring produced	Many	Few
Timing of maturation	Early	Late (with much parental care)
Life expectancy	Short	Long
Lifetime reproductive events	One	More than one
Survivorship curve	Type III	Type I or II

Nest Parasitism

- Parental care is costly, yet often increases offspring survival
 - So get someone else to do it for you
- "Sneaker" males in fish
- Extra-pair copulations in birds
- Specialized brood parasitism in cuckoos has evolved multiple times





- Why do birds feed chicks that aren't their own?
- Parents biased to feeding large offspring
 - Prevents waste of resources on weak or sickly offspring
 - Parasitic eggs tend to hatch earlier than others; grow bigger
 - Cuckoo calls mimic an entire brood of warblers

*The Definitive Guide to the Evolutionary Biology of **Sex***



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"Delightful . . . Easy to understand and hard to resist, it's sex education at its prime—accurate, comprehensive, and hilarious. Your boyfriend will be thrilled. Just don't get too inspired and bite his head off." —*Newsweek*

Olivia Judson



- Boobies typically lay two eggs.
- Both eggs typically do not make it to adulthood.
- What's the benefit to laying two eggs?



Parental Care Game

- Each table has 1 parent. The rest are offspring. Parents take the supply of poker chips. Represents “investment.” That is your investment for entire game.
- Each round, offspring must receive X (1-3) food to survive. Parents may give offspring any amount of food each round.
- Parents score 2 points for every child that survives. Offspring get 2 points for surviving a round, and 1 for each surviving sibling that round.
- Offspring must either “share or scramble.” Divide food equally if sharing. Grab as much as you can from the center if you’re scrambling.